

Start

MERGE FUNCTION (matrix(A) ,first index(p) ,
middle(q) , last index(r)){

$n1 = q - p + 1$

$n2 = r - q$

CREATE left[n1], right[n2]

For i FROM 0 TO (n1 - 1)

left[i] = A[p + i]

For j FROM 0 TO (n2 - 1)

right[j] = A[q + 1 +j]

CREATE i = 0 , j = 0 , k = p

WHILE i < n1 AND j < n2 Do {

IF left[i] <= right[j] THEN

 A[k] = left[i]

 i = i + 1

ELSE

 A[k] = right[j]

 j = j + 1

k = k + 1

}

```

WHILE i < n1 DO
    A[k] = left[i]
    i = i + 1
    k = k + 1
WHILE j < n2 DO
    A[k] = right[j]
    j = j + 1
    k = k + 1
}

```

```

MERGESORT FUNCTION (matrix A , p ,end r) {
    IF p < r THEN
        q = (p + r) / 2

        MERGESORT (A , p , q)
        MERGESORT(A , q + 1 , r)
        MERGE(A , p , q , r)
}

```

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TRIANGLE FUNCTION (A , i){
    IF i >= SIZE(A) - 2 THEN
        RETURN 0

    IF A[i] + A[ i+1 ] > A[ i+2 ]
        RETURN 1

    RETURN TRIANGLE(A , i+1)
}

```

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MAIN {  
    n = NUM OF INDEXES READED  
  
    IF n < 3 THEN  
        Print 0  
        RETURN 0  
  
    CREATE A[n]  
    FOR EACH X IN A DO  
        X = READ  
  
    B = A  
    MERGESORT( B , 0 , n-1 )  
    result = TRIANGLE( B , 0 )  
    PRINT result  
}
```